

THE UNIVERSITY OF AZAD JAMMU & KASHMIR
MUZAFFARABAD

DATABASE MANAGEMENT SYSTEM LAB

Lab 7 — SQL Filters — Advanced Filtering and Assessment
Company & Bookstore Databases

Name	Tayyab Hanif Awan
Roll No	2024-SE-11
Semester	04
Course Title	Database Management System (CS-2204)
Submitted To	Engr. Awais Rathore
Lab No.	07
Tool Used	XAMPP (MySQL / MariaDB) with phpMyAdmin

1. Objective

- Continue practicing advanced filtering techniques on a Company database.
- Apply the same skills to a full standalone graded assessment on a Bookstore database.
- Reinforce combining multiple filter conditions to answer specific business questions.

2. Database Structure

Two databases are used: lab7_company for continued guided practice, and lab7_bookstore for the graded assessment (Q1-Q10).

2.1 lab7_company

Table	Purpose
Employee	Employee records used for the Part B filtering tasks

2.2 lab7_bookstore

Table	Purpose
Book	Book catalog used for the Q1-Q10 graded assessment

3. Key Tasks / Concepts Implemented

- Part B tasks: advanced multi-condition filters on the Employee table.
- Assessment Q1-Q10: filtering, sorting, and pattern-matching queries on the Book table, each carrying assigned marks.
- Verifying each query's output matches the expected business question being asked.

4. Implementation (phpMyAdmin)

All Part B tasks and the Q1-Q10 assessment were executed and verified using phpMyAdmin under XAMPP.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 EmpID	int(11)			No	None			Change Drop More
☐	2 EmpName	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 Gender	char(1)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	4 Salary	decimal(10,2)			Yes	NULL			Change Drop More
☐	5 HireDate	date			Yes	NULL			Change Drop More
☐	6 City	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	7 JobTitle	varchar(40)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	8 DeptName	varchar(40)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Browser	Structure	SQL	Search	Insert	Export	Import	Privileges	Operations	Tracking	More
Employee										
←	→	EmpID	EmpName	Gender	Salary	HireDate	City	JobTitle	DeptName	
☐	Edit Copy Delete	101	Ali Khan	M	120000.00	2018-03-15	Lahore	Senior Engineer	Engineering	
☐	Edit Copy Delete	102	Sara Iqbal	F	95000.00	2019-06-01	Lahore	Software Engineer	Engineering	
☐	Edit Copy Delete	103	Hamza Raza	M	85000.00	2020-01-20	Karachi	Software Engineer	Engineering	
☐	Edit Copy Delete	104	Ayesha Noor	F	110000.00	2017-11-10	Karachi	Marketing Lead	Marketing	
☐	Edit Copy Delete	105	Bilal Ahmed	M	70000.00	2021-04-05	Karachi	Marketing Exec	Marketing	
☐	Edit Copy Delete	106	Fatima Sheikh	F	90000.00	2019-09-12	Islamabad	Accountant	Finance	
☐	Edit Copy Delete	107	Usman Tariq	M	78000.00	2022-02-18	Islamabad	Accountant	Finance	
☐	Edit Copy Delete	108	Maira Javed	F	115000.00	2016-07-22	Lahore	Research Lead	Research	
☐	Edit Copy Delete	109	Zain Abbas	M	60000.00	2023-01-09	Lahore	Research Analyst	Research	
☐	Edit Copy Delete	110	Nida Yousaf	F	72000.00	2022-08-30	NULL	Research Analyst	Research	
☐	Edit Copy Delete	111	Adeel Akhtar	M	88000.00	2020-05-14	Lahore	QA Engineer	Engineering	
☐	Edit Copy Delete	112	Sana Malik	F	102000.00	2018-12-01	Karachi	Sales Manager	Sales	
☐	Edit Copy Delete	113	Talha Hussain	M	65000.00	2023-07-18	Islamabad	Sales Exec	Sales	
☐	Edit Copy Delete	114	Mehwish Anwar	F	80000.00	2021-10-25	Lahore	HR Officer	HR	
☐	Edit Copy Delete	115	Imran Shafi	M	125000.00	2015-04-30	NULL	Director	Engineering	

Structure/Browse tab for employee table in lab7_company database

✓ Showing rows 0 - 5 (6 total, Query took 0.0004 seconds.)

```
SELECT * FROM Employee WHERE HireDate BETWEEN '2020-01-01' AND '2022-12-31';
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: | Filter rows: | Sort by key:

Extra options

				▼ EmpID	EmpName	Gender	Salary	HireDate	City	JobTitle	DeptName	
☐		 Edit	 Copy	 Delete	103	Hamza Raza	M	85000.00	2020-01-20	Karachi	Software Engineer	Engineering
☐		 Edit	 Copy	 Delete	105	Bilal Ahmed	M	70000.00	2021-04-05	Karachi	Marketing Exec	Marketing
☐		 Edit	 Copy	 Delete	107	Usman Tariq	M	78000.00	2022-02-18	Islamabad	Accountant	Finance
☐		 Edit	 Copy	 Delete	110	Nida Yousaf	F	72000.00	2022-08-30	NULL	Research Analyst	Research
☐		 Edit	 Copy	 Delete	111	Adeel Akhtar	M	88000.00	2020-05-14	Lahore	QA Engineer	Engineering
☐		 Edit	 Copy	 Delete	114	Mehwish Anwar	F	80000.00	2021-10-25	Lahore	HR Officer	HR

☐ Check all | With selected: Edit Copy Delete Export

```
SELECT * FROM Employee WHERE Salary NOT BETWEEN 80000 AND 100000;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: | Filter rows: | Sort by key:

Extra options

				EmpID	EmpName	Gender	Salary	HireDate	City	JobTitle	DeptName
☐				101	Ali Khan	M	120000.00	2018-03-15	Lahore	Senior Engineer	Engineering
☐				104	Ayesha Noor	F	110000.00	2017-11-10	Karachi	Marketing Lead	Marketing
☐				105	Bilal Ahmed	M	70000.00	2021-04-05	Karachi	Marketing Exec	Marketing
☐				107	Usman Tariq	M	78000.00	2022-02-18	Islamabad	Accountant	Finance
☐				108	Maira Javed	F	115000.00	2016-07-22	Lahore	Research Lead	Research
☐				109	Zain Abbas	M	60000.00	2023-01-09	Lahore	Research Analyst	Research
☐				110	Nida Yousaf	F	72000.00	2022-08-30	NULL	Research Analyst	Research
☐				112	Sana Malik	F	102000.00	2018-12-01	Karachi	Sales Manager	Sales
☐				113	Talha Hussain	M	65000.00	2023-07-18	Islamabad	Sales Exec	Sales
☐				115	Imran Shafi	M	125000.00	2015-04-30	NULL	Director	Engineering

Console

Outputs of some part B filtering tasks

```
SELECT Title, Price FROM Book WHERE Price > 1500;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

			Title	Price
☐	Edit	Copy	Delete	Kafka on the Shore
☐	Edit	Copy	Delete	Atomic Habits
☐	Edit	Copy	Delete	The Power of Habit
☐	Edit	Copy	Delete	Sapiens

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

```
SELECT Title, Author FROM Book WHERE Title LIKE '%the%';
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

			Title	Author
☐	Edit	Copy	Delete	Kafka on the Shore
☐	Edit	Copy	Delete	The Reluctant Fundamentalist
☐	Edit	Copy	Delete	The Power of Habit

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

```
SELECT * FROM Book WHERE PublishedYear < 2000 AND Price < 1200 ORDER BY Genre ASC, Title ASC;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

			BookID	Title	Author	Genre	Price	StockQty	PublishedYear	Publisher	Language
☐	Edit	Copy	Delete	2	Emma	Jane Austen	Fiction	900.00	8	1815	Penguin
☐	Edit	Copy	Delete	1	Pride and Prejudice	Jane Austen	Fiction	850.00	12	1813	Penguin
☐	Edit	Copy	Delete	14	Raja Gidh	Bano Qudsia	Fiction	900.00	14	1981	Sang-e-Meel
☐	Edit	Copy	Delete	3	Things Fall Apart	Chinua Achebe	Fiction	1100.00	5	1958	Heinemann
☐	Edit	Copy	Delete	12	Rich Dad Poor Dad	Robert Kiyosaki	Finance	1100.00	25	1997	Plata
☐	Edit	Copy	Delete	13	Aab-e-Hayat	Ibn-e-Safi	Mystery	650.00	18	1955	Asrar

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

Some outputs from assessment problem

5. Conclusion

This lab reinforced advanced filtering skills through a full graded assessment, confirming readiness to translate a plain-English business question directly into a correct SQL WHERE clause.